

Morris-Thorne Wormhole Null Geodesics

Daniel T. Murphy

Independent Research • Star-Magic / UQFF Project

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Significance

FIRST wormhole physics integration in the Star Magic / UQFF CP pipeline (CP1–CP4). The wormhole coupling to MUGE resonance gravity via the a_{worm} term is derived in PAPER_375.

Abstract

This paper presents the implementation of Morris-Thorne traversable wormhole null geodesics within the Star Magic UQFF framework. The wormhole metric, geodesic equations, traversal and reflection conditions, and embedding functions are fully specified. This constitutes the first wormhole physics to appear in the Condensed Physics calculator pipeline.

1. Wormhole Metric (Morris-Thorne 1988)

The static, spherically symmetric traversable wormhole metric:

$$ds^2 = -dt^2 + dr^2 + (b^2 + r^2)(d\theta^2 + \sin^2(\theta) d\phi^2)$$

where $b = 1.0$ m is the throat radius. This is the simplest traversable wormhole shape function $b(r) = b_0 = \text{const}$, corresponding to zero radial tidal forces at the throat.

2. Null Geodesic Equations

For null geodesics ($ds^2 = 0$) with conserved energy $E = dt/d\lambda$ and angular momentum $L = (b^2 + r^2) d\phi/d\lambda$:

$$dr/d_\lambda = \pm \sqrt{E^2 - L^2/(b^2 + r^2)}$$

$$d\phi/d_\lambda = L / (b^2 + r^2)$$

$$dt/d_\lambda = E$$

where λ is the affine parameter.

3. Traversal and Reflection Cases

Case 1: Traversal ($L = 0.5$, $E = 1.0$)

$V_{\text{eff}}(r) = L^2/(b^2 + r^2) = 0.25/(1 + r^2)$. Since $V_{\text{eff}} < E^2 = 1$ for all r , the null ray crosses the throat $r = 0$ and continues to the second universe.

Case 2: Reflection ($L = 1.5$, $E = 1.0$)

$V_{\text{eff}}(0) = L^2/b^2 = 2.25 > E^2 = 1$. The turning radius:

$$r_{\min} = \sqrt{L^2/E^2 - b^2} = \sqrt{2.25 - 1} = \sqrt{1.25} = 1.118 \text{ m}$$

Case	L	E	Outcome	r_min (m)
Traversal	0.5	1.0	Passes through throat	0
Reflection	1.5	1.0	Reflects at r_min	~1.118

Table 1. Null geodesic traversal and reflection cases

4. Embedding Functions

The wormhole embedding in Euclidean 3-space (equatorial slice, $\theta = \pi/2$):

$$z_{\text{embed}}(r) = b * \text{arcsinh}(r/b)$$

$$\rho_{\text{embed}}(r) = \sqrt{b^2 + r^2}$$

These define the "funnel" shape visible in standard wormhole visualisations.

5. Numerical Integration Scheme

The C++ propagator uses first-order Euler (upgradeable to RK4):

```
for step = 0..n_steps:
  dr = (+/-sqrt(E^2 - L^2/(b^2+r^2))) * d_lambda
  dphi = (L/(b^2+r^2)) * d_lambda
  dt = E * d_lambda
  r += dr, phi += dphi, t += dt
```

Defaults: $d\lambda = 0.1 \text{ m}$, $n_{\text{steps}} = 100$.

6. Connection to UQFF

The wormhole throat geometry enters the MUGE resonance model via the **wormhole-MUGE coupling term** (PAPER_375):

$$a_{\text{worm}}(r) = f_{\text{worm}} * E_{\text{vac,neb}} / (b^2 + r^2)$$

where $f_{\text{worm}} = 10^{-10}$ (wormhole coupling constant), $E_{\text{vac,neb}} = 7.09 \times 10^{-36} \text{ J}$, $b = 1.0 \text{ m}$, r = evaluation radius. At $r = b = 1 \text{ m}$: $a_{\text{worm}} = 3.545 \times 10^{-46} \text{ m/s}^2$.

7. Implementation

C++: STAR_MAGIC_09SEPT_UQFF_MODULE.cpp, namespace WormholeGeodesics

- Struct: GeodesicState {r, phi, t_coord}
- Functions: drdt(), dphidt(), z_embed(), rho_embed(), propagate(), selftest()

Python: CondensedPhysics4.py, class MorrisThorneWormholeNullGeodesicsCalculator (CP4 #21)

Wolfram macro: WOLFRAM_TERM_WORMHOLE_GEODESIC

References

- [1] Morris, M.S. & Thorne, K.S. (1988). Wormholes in spacetime and their use for interstellar travel. *Am. J. Phys.* 56(5): 395-412.
- [2] Visser, M. (1995). *Lorentzian Wormholes*. Springer-Verlag.